SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A
Debugging & Optimisation Task

Code of Conduct:

I **KONGARA JOGESH(192525035)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Jogesh

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings: IP Address: 192.168.10.65, Subnet Mask: 255.255.255.192 (/26), Default Gateway: 192.168.10.1. However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses: Device A: 2001:db8::ff00:42:8329/64, Device B: 2001:db8:0:0:1::1/64. Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses: Device A: 2001:db8::ff00:42:8329/64, Device B: 2001:db8:0:0:1::1/64. Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.

- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

ANSWER 1: Workstation Configuration Debugging

Given:

- IP Address = **192.168.10.65**
- Subnet Mask = **255.255.255.192 (/26)**
- Default Gateway = **192.168.10.1**

Step 1: Identify the Subnet of the Configured IP Address

Explanation:

The subnet mask **/26** means that the first **26 bits** represent the network portion, while the remaining **6 bits** represent the host portion.

Each /26 subnet contains:

$$2^{(32-26)} = 2^6 = 64 \text{ addresses}$$

Subnet	**Address Range**
1	192.168.10.0 – 192.168.10.63
2	192.168.10.64 – 192.168.10.127
3	192.168.10.128 – 192.168.10.191
4	192.168.10.192 – 192.168.10.255

Step 2: Network Address, Broadcast Address, and Valid Host Range

Subnet	Network Address	Broadcast Address	Valid Host Range
Intended: 192.168.10.0/26	192.168.10.0	192.168.10.63	192.168.10.1 – 192.168.10.62
Actual: 192.168.10.64/26	192.168.10.64	192.168.10.127	192.168.10.65 – 192.168.10.126

Step 3: Verify the Configured IP Address and Gateway

The workstation IP 192.168.10.65 is a valid host address, but only within the 192.168.10.64/26 subnet — it does NOT belong to the intended 192.168.10.0/26 subnet. The configured default gateway, 192.168.10.1, is a valid host address within 192.168.10.0/26, but it does NOT belong to the workstation's actual subnet (192.168.10.64/26). Since a host can only reach a default gateway that resides in its own subnet, the gateway 192.168.10.1 is unreachable from IP 192.168.10.65.

Step 4: Identify the Configuration Error

The root cause of the communication failure is a subnet mismatch between the host IP address and the default gateway. The workstation's IP address (192.168.10.65) places it in the 192.168.10.64/26 subnet, while its configured gateway (192.168.10.1) belongs to the neighbouring 192.168.10.0/26 subnet. Because these two addresses fall on opposite sides of the /26 subnet boundary, the workstation cannot deliver packets to its gateway at Layer 2, and consequently cannot reach any device outside its own (incorrect) subnet — including the intended 192.168.10.0/26 devices.

Step 5: Optimized (Corrected) Configuration

To resolve the fault, the workstation's IP address must be changed to a valid host address within the intended 192.168.10.0/26 subnet, keeping the same mask and gateway:

Parameter	Corrected Value
IP Address	192.168.10.10 (any address in 192.168.10.1 – 192.168.10.62)
Subnet Mask	255.255.255.192 (/26)
Default Gateway	192.168.10.1
Network Address	192.168.10.0
Broadcast Address	192.168.10.63

Step 6: Usable Host Addresses After Correction

A /26 subnet provides $2^{(32-26)} = 64$ total addresses, of which 2 are reserved (network and broadcast). The number of usable host addresses after correction is therefore $64 - 2 = 62$, spanning 192.168.10.1 to 192.168.10.62. With the workstation now correctly addressed at 192.168.10.10 with gateway 192.168.10.1, it resides in the same broadcast domain as the rest of the 192.168.10.0/26 subnet and can communicate normally.

Final Answer

Item	Answer
Subnet	192.168.10.64/26
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Valid Host Range	192.168.10.65 – 192.168.10.126
Configured IP	Valid
Configured Gateway	Invalid
Correct Gateway	192.168.10.126 (example)
Usable Hosts	62

ANSWER 2: IPv6 Configuration Debugging (Devices A & B)

Given:

- **Device A = 2001:db8::ff00:42:8329/64**
- **Device B = 2001:db8:0:0:1::1/64**

Step 1: Expand Both IPv6 Addresses

Explanation:
IPv6 addresses use "::" to represent consecutive zeros. Expand each address into its full 8-block notation.

Device A:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Device B:

```
2001:0db8:0000:0000:0001:0000:0000:0001
```

Step 2: Determine the Network Prefix of Each Device (/64)

With a /64 prefix, the first 64 bits (first four hextets) define the network portion, and the last 64 bits form the interface identifier.

Device	Full Address	Network Prefix (/64)
Device A	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:0db8:0000:0000::/64
Device B	2001:0db8:0000:0000:0001:0000:0000:0001	2001:0db8:0000:0000::/64

Step 3: Verify Whether Both Devices Belong to the Same Subnet

Comparing the first four hextets of both addresses — 2001:0db8:0000:0000 for Device A and 2001:0db8:0000:0000 for Device B — shows they are identical. Both devices therefore share the exact same network prefix, 2001:db8::/64, and are correctly placed in the same subnet at the addressing level.

Step 4: Identify the Addressing / Prefix Mismatch

Unlike a typical VLSM/prefix-mismatch fault, this analysis shows there is no prefix mismatch between Device A and Device B — both resolve to the same /64 network (2001:db8::/64), and their interface identifiers (...ff00:0042:8329 for A and ...0001:0000:0000:0001 for B) are distinct, so there is no duplicate-address conflict either. Since the addressing itself is technically correct, the communication failure must be attributed to a cause outside of the IPv6 prefix, such as: a Neighbor Discovery Protocol (NDP) failure, a firewall or Access Control List blocking ICMPv6 traffic, the two hosts sitting on different VLANs or physical segments despite sharing a logical /64, or Duplicate Address Detection (DAD) issues during interface configuration.

Step 5: Optimized IPv6 Configuration

Although the prefix itself is not at fault, best practice for clarity and easier troubleshooting is to assign both devices short, unambiguous addresses within the confirmed shared subnet, and to verify Layer 2 connectivity (same VLAN/switch port group) alongside the addressing:

Device	Optimized Address
Device A	2001:db8::1/64
Device B	2001:db8::2/64

Step 6: Available Host Addresses Within the Corrected Subnet

For a /64 prefix, 128 − 64 = 64 bits remain for host addressing, giving 2^64 = 18,446,744,073,709,551,616 possible addresses in the 2001:db8::/64 subnet — effectively unlimited for any practical LAN, confirming that address space is not a limiting factor in this scenario.

Final Answer

- Expanded Address A = **2001:0db8:0000:0000:0000:ff00:0042:8329**
- Expanded Address B = **2001:0db8:0000:0000:0001:0000:0000:0001**
- Network Prefix = **2001:db8:0:0::/64**
- Same Subnet = **Yes**
- Optimized Addresses = **2001:db8::10**, **2001:db8::20**
- Available Hosts = $\mathbf{2^{64}}$

ANSWER 3: Optimizing Department Subnet Allocation (192.168.1.0/24)

Given:

Network = 192.168.1.0/24

Subnet Masks:

- **/26**
- **/27**
- **/28**

Step 1: Original Allocation — Network, Broadcast, and Host Range

Allocating sequentially from the start of the block in the given mask order (/26, /27, /28) for three departments (Sales, IT, HR):

Department	Subnet	Network Addr.	Broadcast Addr.	Host Range	Usable Hosts
Sales	192.168.1.0/26	192.168.1.0	192.168.1.63	.1 – .62	62
IT	192.168.1.64/27	192.168.1.64	192.168.1.95	.65 – .94	30
HR	192.168.1.96/28	192.168.1.96	192.168.1.111	.97 – .110	14

Step 2: Compare Allocation Against Actual Host Requirements

Suppose the true device counts are: Sales needs 55 hosts, IT needs 20 hosts, and HR needs 8 hosts. Comparing these against the capacity of each assigned subnet reveals the inefficiency:

Department	Hosts Needed	Assigned Subnet	Capacity	Result
Sales	55	/26 (192.168.1.0)	62 usable	Efficient — fits with small margin
IT	20	/27 (192.168.1.64)	30 usable	Oversized — 10 addresses wasted
HR	8	/28 (192.168.1.96)	14 usable	Oversized, yet still the tightest fit

Step 3: Identify Inefficient / Incorrect Allocations

The core problem reported — some departments short of addresses, others with large unused pools — appears when a department needing more hosts than /28 allows (14 usable) is mistakenly assigned a /28, or when a small department is mistakenly given a /26. For example, if HR (needing only 8 hosts) had instead been assigned the /26 block, 54 of its 62 addresses would sit permanently unused; and if a growing department needing 20+ hosts had been squeezed into a /28, , forcing awkward re-addressing later. This mismatch between subnet size and actual demand is the fundamental debugging finding: subnets were assigned by a fixed mask list rather than by measured host requirements. Additionally, 256 − (64+32+16) = 144 addresses (over half the /24 block) remain completely unallocated,

Step 4: Debugged / Optimized VLSM Allocation Plan

Applying VLSM correctly — sorting departments from largest to smallest actual requirement and assigning the smallest subnet that satisfies each — produces a tightly-fitted, wastage-minimized plan:

Department	Hosts Needed	Optimized Subnet	Network Addr.	Broadcast Addr.	Usable Hosts
Sales	55	/26	192.168.1.0	192.168.1.63	62
IT	20	/27	192.168.1.64	192.168.1.95	30
HR	8	/28	192.168.1.96	192.168.1.111	14

Step 5: Conclusion

In this corrected plan, every department's subnet is the smallest standard block that still comfortably covers its real host count, eliminating both the "insufficient addresses" complaint (by upsizing under-provisioned departments to the next larger mask) and the "large unused pool" complaint (by downsizing over-provisioned departments to the next smaller mask). The remaining 192.168.1.112 – 192.168.1.255 range (144 addresses) is left as a clearly identified,

Final Answer

Subnet	Network	Broadcast	Host Range	Usable Hosts
/26	192.168.1.0	192.168.1.63	192.168.1.1–192.168.1.62	62
/27	192.168.1.64	192.168.1.95	192.168.1.65–192.168.1.94	30
/28	192.168.1.96	192.168.1.111	192.168.1.97–192.168.1.110	14

ANSWER 4: IPv6 Configuration Debugging (Devices A & B) — Repeat Scenario

Step : Expand Both IPv6 Addresses

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Step 1: Expand Both Addresses to Complete Notation

Device A — 2001:db8::ff00:42:8329 has 5 explicit groups (2001, db8, ff00, 42, 8329), so the "::" expands to fill the remaining 8 − 5 = 3 groups with zeros:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B — 2001:db8:0:0:1::1 has 6 explicit groups (2001, db8, 0, 0, 1, 1), so the "::" expands to fill the remaining 8 − 6 = 2 groups with zeros:

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Determine the Network Prefix of Each Device (/64)

With a /64 prefix, the first 64 bits (first four hextets) define the network portion, and the last 64 bits form the interface identifier.

Device	Full Address	Network Prefix (/64)
Device A	2001:0db8:0000:0000:0000:ff00:0042:8329	2001:0db8:0000:0000::/64
Device B	2001:0db8:0000:0000:0001:0000:0000:0001	2001:0db8:0000:0000::/64

Step 3: Verify Whether Both Devices Belong to the Same Subnet

Comparing the first four hextets of both addresses — 2001:0db8:0000:0000 for Device A and 2001:0db8:0000:0000 for Device B — shows they are identical. Both devices therefore share the exact same network prefix, 2001:db8::/64, and are correctly placed in the same subnet at the addressing level.

Step 4: Identify the Addressing / Prefix Mismatch

Unlike a typical VLSM/prefix-mismatch fault, this analysis shows there is no prefix mismatch between Device A and Device B — both resolve to the same /64 network (2001:db8::/64), and their interface identifiers (...ff00:0042:8329 for A and ...0001:0000:0000:0001 for B) are distinct, so there is no duplicate-address conflict either. Since the addressing itself is technically correct, the communication failure must be attributed to a cause outside of the IPv6 prefix, such as: a Neighbor Discovery Protocol (NDP) failure, a firewall or Access Control List blocking ICMPv6 traffic, the two hosts sitting on different VLANs or physical segments despite sharing a logical /64, or Duplicate Address Detection (DAD) issues during interface configuration.

Step 5: Optimized IPv6 Configuration

Although the prefix itself is not at fault, best practice for clarity and easier troubleshooting is to assign both devices short, unambiguous addresses within the confirmed shared subnet, and to verify Layer 2 connectivity (same VLAN/switch port group) alongside the addressing:

Device	Optimized Address
Device A	2001:db8::1/64
Device B	2001:db8::2/64

Step 6: Available Host Addresses Within the Corrected Subnet

For a /64 prefix, 128 − 64 = 64 bits remain for host addressing, giving 2^64 = 18,446,744,073,709,551,616 possible addresses in the 2001:db8::/64 subnet — effectively unlimited for any practical LAN, confirming that address space is not a limiting factor in this scenario.

Final Answer

- Same Network = **Yes**
- Network Prefix = **2001:db8:0:0::/64**
- Corrected Addresses = **2001:db8::10** and **2001:db8::20**
- Available Hosts = **2^{64}**

ANSWER 5: Routing Table Debugging

Given Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP = **192.168.2.100**

Step 1: Matching Routing Entry via Longest-Prefix Match

Applying the /24 mask to 192.168.2.100 gives a network portion of 192.168.2.0. Checking this against the table: 192.168.1.0/24 does not match (differs in the third octet); 192.168.2.0/24 matches exactly. Since this is the only specific-prefix match (and it is more specific than the 0.0.0.0/0 default), it is selected by longest-prefix match.

Routing Table Entry	Next Hop	Match for 192.168.2.100?
192.168.1.0/24	10.0.0.2	No
192.168.2.0/24	10.0.0.3	Yes — selected
Default Route (0.0.0.0/0)	10.0.0.1	Fallback only

Step 2: Next-Hop Router Selected for Forwarding

Based on the matched entry 192.168.2.0/24, the router forwards the packet to next hop 10.0.0.3. This is the router interface/neighbour expected to deliver the packet onward to the 192.168.2.0/24 network.

Step 3: Verify the Destination Belongs to the Intended Subnet

Under a /24 mask, 192.168.2.0/24 spans network address 192.168.2.0, broadcast address 192.168.2.255, and a valid host range of 192.168.2.1 – 192.168.2.254. The destination 192.168.2.100 falls within this valid host range, confirming it is a legitimate address inside the intended subnet — the routing table entry itself is correctly scoped.

Step 4: Identify the Routing Configuration Error

Since the longest-prefix match logic correctly selects 192.168.2.0/24 → 10.0.0.3, and 192.168.2.100 is a valid host within that subnet, the routing table's forwarding decision is not logically wrong. The failure to reach the destination therefore points to a downstream fault rather than a table-matching error — most likely one of the following: (a) the next-hop 10.0.0.3 is unreachable, down, or has an interface/link failure; (b) 10.0.0.3 does not have a return route back toward the source network, causing silent packet loss (asymmetric routing); (c) 10.0.0.3 is incorrectly configured as the next hop when the actual gateway to 192.168.2.0/24 is a different address; or (d) an access-control list or firewall on the path to 10.0.0.3 is silently dropping traffic to 192.168.2.100.

Step 5: Optimizing the Routing Table

- Verify reachability of next hop 10.0.0.3 (ping/traceroute) before trusting the static entry.
- Confirm 10.0.0.3 has a return route back to the source subnet to avoid asymmetric routing drops.
- Add a floating/backup static route (higher administrative distance) via an alternate next hop for 192.168.2.0/24, so a single next-hop failure does not fully isolate the subnet.
- Enable route monitoring (e.g., IP SLA / periodic reachability checks) to automatically withdraw the 10.0.0.3 route if it becomes unreachable, falling back to the default route or backup path.
- Double-check the default route (0.0.0.0/0 via 10.0.0.1) is present and correctly configured as the safety-net path for any traffic that fails to match a specific entry.

Step 6: Route Used When No Matching Destination Exists

If an incoming packet's destination does not match either 192.168.1.0/24 or 192.168.2.0/24, the router falls back on the Default Route, forwarding the packet to next hop 10.0.0.1. This ensures traffic to any unlisted destination is still forwarded rather than dropped, provided 10.0.0.1 itself leads toward a valid onward path (e.g., the ISP or core network).

Final Answer

- Matching Route = **192.168.2.0/24**
- Next Hop = **10.0.0.3**
- Destination IP = **Valid**
- Possible Error = **Incorrect next hop or interface problem**
- Optimized Route = **192.168.2.0/24 → 10.0.0.3**

- Default Route = **10.0.0.1**